

C Tasks

Week 2

Q1: 1- Write a function that swaps two numbers using pointers.

2- Write a function that reverses a string using pointers.

Q2: 1- Write a function that takes an array of integers and its size as parameters and returns the maximum and minimum elements and average (numbers may be zero, negative, positive).

2- Write a program to take size of array of integers and get their values and a function to reverse this array without using another array.

Q3: Write a function that takes array of integers and sorts it using Bubble Sort algorithm. (you should use function swap that you write in first question in point 1).

Q4: 1- Write a program to take rows and columns of a matrix of integers as input from the user and matrix values and print it and function that finds the maximum and minimum element and average of a 2D array. (numbers may be zero, negative, positive).

2- Write a function to take 2D array of integers and return array of the sum of every row and the maximum sum of row in this 2D array. Hint (you may use call by reference, you must now how to pass 2d array to functions)

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows : 2
enter number of columns : 2
enter the element in 1th row , 1th column : 1
enter the element in 1th row , 2th column : 2
enter the element in 2th row , 1th column : 3
enter the element in 2th row , 2th column : 4
the matrix is
1 2
3 4
maximum value is : 4
minimum value is : 1
average is : 2.50

Process returned 0 (0x0)   execution time : 4.746 s
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows : 2
enter number of columns : 2
enter the element in 1th row , 1th column : -1
enter the element in 1th row , 2th column : 4
enter the element in 2th row , 1th column : 2
enter the element in 2th row , 2th column : -7
the matrix is
-1 4
2 -7
the sum of rows is
the sum of 1th row is : 3
the sum of 2th row is : -5
the maximum sum of row is : 3
Process returned 0 (0x0)   execution time : 12.283 s
Press any key to continue.
```

Q5: 1-What is a string in C?

2-What is the null terminator (`\0`) in a string? Why is it necessary?

3-Write a program to concatenate two strings using a loop. Without using(`strcat()`) .

4-Write a function to copy one string into another without using `strcpy()`.

5- Write a function that checks whether a given string is a palindrome.

A palindrome is a word, phrase, number, or any sequence of characters that reads the same forward and backward (ignoring spaces, punctuation, and capitalization).

For example: Words: "madam" , "racecar" , "level"

Phrases: "A man, a plan, a canal, Panama" , "Was it a car or a cat I saw?"

Numbers: 121 , 12321

6-Write a function to remove all whitespace from a string and replace them with '`_`'

Input: "This is a test string!" output: "This_is_a_test_string!"

Bonus :

Q1: Write a function to split a string by a delimiter in C. Hint (you may use `strtok()`).

Q2: Write a function to find the transpose of a given matrix. (convert rows to columns and columns to rows). Hint (you may use dynamic allocation and return pointer to pointer of int from function).

Q3: Write a function that checks if a given matrix is symmetric with its transpose. Hint (i.e., $A[i][j] == A[j][i]$).

Q4: Write a C program to multiply two matrices.

Show their matrix multiplication techniques and rules:

<https://www.mathsisfun.com/algebra/matrix-multiplying.html>

Outputs of Q4:

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows of first matrix : 2
enter number of columns of first matrix : 3
enter the element in 1th row , 1th column of first matrix : 1
enter the element in 1th row , 2th column of first matrix : 2
enter the element in 1th row , 3th column of first matrix : 3
enter the element in 2th row , 1th column of first matrix : 4
enter the element in 2th row , 2th column of first matrix : 5
enter the element in 2th row , 3th column of first matrix : 6
the matrix is
1 2 3
4 5 6
enter number of rows of second matrix : 3
enter number of columns of second matrix : 2
enter the element in 1th row , 1th column of second matrix : 7
enter the element in 1th row , 2th column of second matrix : 8
enter the element in 2th row , 1th column of second matrix : 9
enter the element in 2th row , 2th column of second matrix : 10
enter the element in 3th row , 1th column of second matrix : 11
enter the element in 3th row , 2th column of second matrix : 12
the matrix is
7 8
9 10
11 12
its multiplication:
58 64
139 154
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows of first matrix : 1
enter number of columns of first matrix : 2
enter the element in 1th row , 1th column of first matrix : 1
enter the element in 1th row , 2th column of first matrix : 2
the matrix is
1 2
enter number of rows of second matrix : 3
enter number of columns of second matrix : 1
enter the element in 1th row , 1th column of second matrix : 3
enter the element in 2th row , 1th column of second matrix : 4
enter the element in 3th row , 1th column of second matrix : 5
the matrix is
3
4
5
they can't be multiplied with each other
Process returned 0 (0x0)   execution time : 8.669 s
Press any key to continue.
```

Output samples of Q2 and Q3:

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows : 2
enter number of columns : 2
enter the element in 1th row , 1th column : 2
enter the element in 1th row , 2th column : 3
enter the element in 2th row , 1th column : 4
enter the element in 2th row , 2th column : 2
the matrix is
2 3
4 2
its transpose
2 4
3 2
the matrix is not symmetric with its transpose
Process returned 0 (0x0)   execution time : 12.747 s
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows : 3
enter number of columns : 1
enter the element in 1th row , 1th column : 1
enter the element in 2th row , 1th column : 2
enter the element in 3th row , 1th column : 3
the matrix is
1
2
3
its transpose
1 2 3
the matrix is not symmetric with its transpose
Process returned 0 (0x0)   execution time : 11.452 s
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows : 2
enter number of columns : 2
enter the element in 1th row , 1th column : 2
enter the element in 1th row , 2th column : 4
enter the element in 2th row , 1th column : 4
enter the element in 2th row , 2th column : 2
the matrix is
2 4
4 2
its transpose
2 4
4 2
the matrix is symmetric with its transpose
Process returned 0 (0x0)   execution time : 15.312 s
Press any key to continue.
```